Polling vs Long-Polling vs Webhooks

Polling, long-polling, and webhooks are three techniques used in applications for getting updates or information, each with its own mechanism and use case.

Polling

- **Definition**: Polling is a technique where the client repeatedly requests (polls) a server at regular intervals to get new or updated data.
- **Characteristics**:
  - **Regular Requests**: The client makes requests at fixed intervals (e.g., every 5 seconds).
  - **Client-Initiated**: The client initiates each request.
- **Example**: A weather app that checks for updated weather information every 15 minutes by sending a request to the weather server.
- **Pros**:
  - **Simple to Implement**: Easy to set up on the client side.
- **Cons**:
  - **Inefficient**: Generates a lot of unnecessary traffic and server load, especially if there are no new updates.
  - **Delay in Updates**: There's always a delay between the actual update and the client receiving it.

Long-Polling

- **Definition**: Long-polling is an enhanced version of polling where the server holds the request open until new data is available to send back to the client.
- **Characteristics**:
  - **Open Connection**: The server keeps the connection open for a period until there's new data or a timeout occurs.
  - **Reduced Traffic**: Less frequent requests compared to traditional polling.
- **Example**: A chat application where the client sends a request to the server and the server holds the request until new messages are available. Once new messages arrive, the server responds, and the client immediately sends another request.
- **Pros**:
  - **More Timely Updates**: Clients can receive updates more quickly after they occur.
  - **Reduced Network Traffic**: Less frequent requests than standard polling.
- **Cons**:
  - **Resource Intensive on the Server**: Holding connections open can consume server resources.

Webhooks

- **Definition**: Webhooks are user-defined HTTP callbacks that are triggered by specific events. When the event occurs, the source site makes an HTTP request to the URL configured for the webhook.
- **Characteristics**:
  - **Server-Initiated**: The server sends data when there's a new update, without the client needing to request it.
  - **Event-Driven**: Triggered by specific events in the server.
- **Example**: A project management tool where a webhook is set up to notify a team's chat application whenever a new task is created. The creation of the task triggers a webhook that sends data directly to the chat app.
- **Pros**:
  - **Real-Time**: Provides real-time updates.
  - **Efficient**: Eliminates the need for polling, reducing network traffic and load.
- **Cons**:
  - **Complexity in Handling**: The client needs to be capable of receiving and handling incoming HTTP requests.
  - **Security Considerations**: Requires secure handling to prevent malicious data reception.

Key Differences

- **Initiation and Traffic**: Polling is client-initiated with frequent traffic, long-polling also starts with the client but reduces traffic by keeping the request open, and webhooks are server-initiated, requiring no polling.
- **Real-Time Updates**: Webhooks offer the most real-time updates, while polling and long-polling have inherent delays.

Conclusion

The choice between polling, long-polling, and webhooks depends on the application's requirements for real-time updates, server and client capabilities, and efficiency considerations. Polling is simple but can be inefficient, long-polling offers a middle ground with more timely updates, and webhooks provide real-time updates efficiently but require the client to handle incoming requests.